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Wireless detonator arrangement

Abstract

A wireless detonator arrangement wherein an initiator is positioned between a power supply and a housing which contains an explosive and through which extends conductors which connect a signal processor to the power supply and to the initiator.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase application of International Patent Application No. PCT/ZA2022/050029 filed Jul. 7, 2022 which claims priority to South Africa Patent Application ZA 2021/06078 filed Aug. 24, 2021, the entire contents of both of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION

(2) This invention relates to a wireless detonator arrangement.

(3) An arrangement of the aforementioned kind can be a bulky device. The arrangement requires a receiver which can receive electromagnetic command signals which travel, at times, through rock. An electrical supply is called for to power the receiver. Provision must be made for a processor to implement command signals relating to synchronisation and firing. An initiator, which may be in the form of a suitably configured detonator, is also required. Electrical energy is needed to ignite the initiator.

(4) Certain applications also call for the detonator arrangement to be coupled in an effective manner to a booster which contains a secondary explosive.

(5) To address the challenges relating to storage and transport a modular approach may be required so that hazardous components can be separated from non-hazardous components. This aspect mitigates against assembly of the detonator arrangement under factory conditions and, conversely, means that assembly on site by an operator, at a blast bench, should be readily effected in a safe and efficient way.

(6) An object of the invention is to address the aforementioned requirement.

SUMMARY OF THE INVENTION

(7) The invention provides a wireless detonator arrangement which includes an initiator, a first housing in which is mounted a power supply, a second housing which contains an explosive material, and a third housing which is engageable with a first end of the second housing and in which is mounted a signal processing assembly which includes a receiver and a processor for processing wireless signals received by the receiver, wherein the first housing is engageable with a second end of the second housing thereby to connect the power supply to the signal processing assembly via a conductive path which extends through the second housing from the first housing to the third housing, and wherein the signal processing assembly, responsive to a wireless fire command signal from a blast controller, causes ignition of the initiator.

(8) Preferably the initiator is engageable with or is mounted in or to, at least the second housing.

(9) The third housing may include a switch which is actuatable manually or electrically to connect the power supply to the signal processing assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention is further described by way of example with reference to the accompanying drawings in which:

(2) FIG. 1 shows a wireless detonator arrangement, according to the invention, in a disassembled state, and

(3) FIG. 2 depicts the wireless detonator arrangement of FIG. 1, fully assembled.

DESCRIPTION OF PREFERRED EMBODIMENT

(4) FIG. 1 of the accompanying drawings illustrates a wireless detonator arrangement **10**, according to the invention, in a disassembled configuration. FIG. 2 shows the detonator arrangement **10** fully assembled.

(5) The detonator arrangement **10** includes a first housing **12**, a second housing **14** and a third housing **16**.

- (6) Each housing is tubular and, depending on volume requirements, may be elongate.
- (7) The first housing **12** contains a battery power supply **18**.
- (8) The second housing **14** contains a secondary explosive **20** and constitutes a booster. The second housing **14** has a first end **22** and a second end **24** in which is formed a recess **26** which faces a recess **26A** in an end **28** of the first housing **12**.
- (9) Mounted inside the third housing **16** is a signal processing assembly **30** which includes an electromagnetic communication signal receiver **32** which is coupled to a receiving antenna **34**, and to a signal processor **36**.
- (10) The arrangement **10** also includes an initiator **40** which, optionally, is mounted inside a casing **42**.
- (11) The initiator **40** and the housing **14**, which contains the explosive material **20**, are hazardous goods. If these components are not connected to the first housing **12** nor to the third housing **16** then the housings **12** and **16** are not regarded as hazardous goods and can be transported and stored without taking into account regulations which apply to hazardous goods.
- (12) On site for assembly of the wireless detonator arrangement **10** an end of the initiator **40** is inserted into the recess **26**. The first housing **12** is then physically engaged with the second housing **14** with a screw action which couples threads (not shown) on the first housing **12** with complementary threads on the second housing **14**. An opposing end of the initiator then fits into the recess **26A**. As an alternative to the screw action a bayonet-type fitting may be used to couple the housings physically to one another.
- (13) When the housings **12** and **14** are so coupled electrical contacts **46** on the first housing **12**, which are electrically connected to the power supply **18**, are brought into electrical connection with contacts **48** at the second end **24** of the second housing. These contacts **48** are electrically connected to conductors **50** which extend from the second end **24** to the first end **22** and which terminate in contacts **52** at the first end. The third housing **16** is similarly connected to the second housing **14**, again using a thread action or a bayonet-type fitting. The contacts **52** are thereby then electrically connected to contacts **54** in the third housing **16**. Electrical conductors **56** connect the contacts **54** to the signal processing assembly **30**. Thus a conductive path is directly established from the power supply **18** to the signal processing assembly **30**. To enhance and simplify electrical connectivity the contacts **48** and **52**, say, could be of circular form i.e. similar to slip rings.
- (14) The fully assembled wireless detonator **10** is shown in FIG. 2.
- (15) A switch **64** which, in this embodiment, is on the third housing **16**, is used to prevent the power source **18** from energising any of the electrical components until such time as operation is required, at site. As needed, at the time of deployment of the arrangement **10** at a blast bench, the switch **64** is turned on so that the power source **18** can then energise the receiver **32** and the processor **36** which, in turn, operating in response to a blast control signal, are used to trigger the initiator **40**.
- (16) The switch **64** could be electronically controlled in that it is only operable when a signal, which preferably is encoded, from an external source, such as a blast controller **68**, is received by the receiver **32**. Thereafter the receiver **32** and the processor **36**, using energy harvesting techniques whereby energy from a signal from the blast controller which is induced into coils (not shown), actuate the switch **64** and the power source **18** is then electrically connected to the processor **36** and to the receiver **32**.
- (17) The housing **12** and the housing **16** have formations **70** and **72** at their respective free ends which can be used to suspend the assembled wireless detonator arrangement **10** from a cord, which goes to surface, at a desired location in a borehole.
- (18) In use of the initiating arrangement **10** a wireless command signal sent from the blast controller **68**, which is at a remote location relative to the arrangement **10**, is received by the antenna **34** and passed to the receiver **32**. The processor **36** implements in a known manner any instruction contained in the command signal. The initiating arrangement **10** is usually first brought

into synchronism with other similar initiating arrangements (not shown) in a blast system. Thereafter when a fire command signal is received by the receiver **32**, the processor **36**, after a pre-programmed timing delay interval, transmits a fire signal to the initiator **40** which causes ignition thereof. That in turn causes ignition of the explosive material **20**.

(19) Power for the sequence of operations is derived from the power supply **18** which energises the receiver **32** and the processor **36** through a conductive path formed by the abutting contacts **46** and **48**, and **52** and **54**, and the conductors **50** and **56**.

(20) Energy from the power supply **18** can be stored in one or more capacitors **76** in the initiator **40** or in the processor **36** or, possibly, in the initiator **40** and in the processor **36**. This enables the initiator **40** and the processor **36** to function independently of the power supply **18** once a fire signal has been received e.g. to implement a programmed time delay and then to fire the detonator. This is beneficial in that a shock wave produced by a previously fired explosive could interrupt the connection of the power supply **18** to the processor **36**.

(21) Assembly of the arrangement on site at a blast bench is readily effected as the components are easily connected to one another. These components can only be connected in the manner shown—any attempt to connect the components to one another in a different way would not be successful.

(22) The detonator arrangement **10** allows for use to be made of a substantially standard detonator in the initiator **40**.

(23) The initiator **40** and the housing **14**, with the explosive **20**, are kept separate from the first housing **12** and the third housing **16** until such time as on site assembly is required. This facilitates storage and transport requirements.

(24) In a different system (FIG. 2) the arrangement **10** is suspended in a borehole from conductors **78** which are connected to a top-box **80** at a mouth of the borehole. The top-box **80** can communicate wirelessly with a blast controller, and via the conductors **78** with the arrangement **10**, to control the blasting process.

Claims

1. A wireless detonator arrangement (**10**) which includes an initiator (**40**), a first housing (**12**) in which is mounted a power supply (**18**), and contacts (**46**) connected to the power supply (**18**), a second housing (**14**) which contains an explosive material (**20**), contacts (**52**) at a first end (**22**) of the second housing (**14**), contacts (**48**) at a second end (**24**) of the second housing (**14**), and conductors (**50**) which extend from the contacts (**48**) to the contacts (**52**), and a third housing (**16**) which is engageable with the first end (**22**) of the second housing (**14**), a signal processing assembly (**30**) mounted inside the third housing (**16**), the signal processing assembly (**30**) including a receiver (**32**) and a processor (**36**) for processing wireless signals received by the receiver (**32**), and contacts (**54**) connected to the signal processing assembly (**30**), the first housing (**12**) being engageable with the second end (**24**) of the second housing (**14**) thereby to connect the power supply (**18**) to the signal processing assembly (**30**) via a conductive path comprising the contacts (**46**, **48**), the conductors (**50**) and the contacts (**52**, **54**), and wherein the signal processing assembly (**30**), responsive to a fire command signal, causes ignition of the initiator (**40**).
2. The wireless detonator arrangement (**10**) according to claim 1 wherein the initiator (**40**) is engageable with or is mounted in or to at least the second housing (**14**).
3. The wireless detonator arrangement (**10**) according to claim 1 wherein the third housing (**16**) includes a switch (**64**) which is actuable manually or electrically to connect the power supply (**18**) to the signal processing assembly (**30**).
4. The wireless detonator initiating arrangement (**10**) according to claim 1 which includes a capacitor (**76**) which is in at least one of the signal processing assembly (**30**) and the initiator (**40**).

and which is operable to enable a firing sequence to be executed independently of the power supply (18) once said fire command signal has been received by the signal processing assembly (30).
